ECE 329 **Homework 4 — Solutions**

1.

a) (3 pts)Using $\mathbf{E} = -\nabla V$, the electric field $\mathbf{E}$ at $x = \frac{d}{2}$ is calculated as

$$\begin{aligned}\mathbf{E}(x) &= -V_a \frac{\partial\,(x/d)^{4/3}}{\partial x}\hat{x} \\ &= -4\frac{x^{1/3}}{d^{4/3}}\hat{x}\,\Big|_{x=\frac{d}{2}} \\ &= -\frac{2^{\frac{5}{3}}}{d}\hat{x}\,\frac{V}{\mathrm{m}}.\end{aligned}$$

b) (3 pts)Using Gauss's law $\rho = \nabla \cdot \mathbf{D}$, the volumetric free-charge density at $x = \frac{d}{4}$ is written

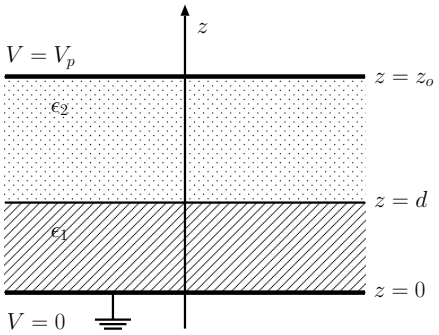
$$\begin{aligned}\rho(x) &= \epsilon_o \nabla \cdot \mathbf{E} \\ &= -4\epsilon_o \frac{1}{d^{4/3}}\left(\frac{\partial x^{1/3}}{\partial x}\right)\Big|_{x=\frac{d}{4}} \\ &= -\frac{4^{\frac{5}{3}}}{3d^2}\epsilon_o\,\frac{\mathrm{C}}{\mathrm{m}^3}.\end{aligned}$$

c) (4 pts)The surface charge density on the anode is given by

$$\begin{aligned}\rho_s &= \hat{n} \cdot (\mathbf{D}^+ - \mathbf{D}^-) \\ &= 4\epsilon_o \frac{x^{1/3}}{d^{4/3}}\hat{x}\,|_{x=d} \\ &= \frac{4}{d}\epsilon_o\,\frac{\mathrm{C}}{\mathrm{m}^2},\end{aligned}$$

where $\hat{n} = -\hat{x}$.

2. The problem can be skecthed as shown in the following figure.



a) (5 pts)For some potential $V_p > 0$, the electric potential increase along $+\hat{z}$ direction with different rates in $0 \le z \le d$ and $d \le z \le z_0$regions. Following the similar procedures in problem 2 and 3, we find

$$\mathbf{E}(z) = \begin{cases} \frac{V_d - 0}{d}(-\hat{z}), & 0 \le z \le d \\ \frac{V_p - V_d}{z_0 - d}(-\hat{z}), & d \le z \le z_0 \end{cases},$$

and

$$\mathbf{D}(z) = \begin{cases} \epsilon_1 \frac{V_d - 0}{d}(-\hat{z}), & 0 \le z \le d \\ \epsilon_2 \frac{V_p - V_d}{z_0 - d}(-\hat{z}), & d \le z \le z_0 \end{cases}.$$

For dielectric interface $\rho_s = 0$, we match the boundary $\mathbf{D}(z)$ fields at $z = d$ interface and obtain

$$\epsilon_1 \frac{V_d - 0}{d} = \epsilon_2 \frac{V_p - V_d}{z_0 - d}$$

$$\Longrightarrow V_d = V_p \frac{d\epsilon_2}{d\epsilon_2 + (z_0 - d)\epsilon_1} (\hat{z})\,\mathrm{V}.$$

Given V_d the electric potential everywhere can be found in terms of d, ϵ_1, ϵ_2, and z_0

$$\mathbf{V}(z) = \begin{cases} V_d \frac{z}{d}, & 0 \le z \le d \\ V_d(1 + K(z - d)), & d \le z \le z_0 \end{cases}$$

where $K = \frac{d\epsilon_2}{(d - z_0)(d\epsilon_2 + (z_0 - 3)\epsilon_1)}$.

b) (5 pts)Using boundary condition of displacement fields at $z = 0$ PEC plate, we find

$$\rho_{z=0} = \frac{v_d}{d} \epsilon_1 (-\hat{z}) = -\frac{18}{5} \epsilon_0 \frac{\mathrm{C}}{\mathrm{m}^2}.$$

And naturally, the surface charge density at $z = z_0$ PEC plate has to be

$$\rho_{z=z_0} = \frac{18}{5} \epsilon_0 \frac{\mathrm{C}}{\mathrm{m}^2}$$

such that the electric potential for $z < 0$ and $z > z_0$is zero.

3.

a) (2 pts)In vacuum, the displacement vector is $\mathbf{D} = \epsilon_o \mathbf{E} + \mathbf{P}$. Thus, the displacement field between the plates is

$$\mathbf{D} = 4\epsilon_o \hat{z} \frac{\mathrm{C}}{\mathrm{m}^2},$$

from which we obtain the polarization field as $\mathbf{P} = 0 \frac{\mathrm{C}}{\mathrm{m}^2}$.

b) (2 pts)Since $\mathbf{D} = \mathbf{0}$ for $z < 0$, i.e. $\mathbf{D}\,|_{z=0^-}$ the surface charge on the plate at $z = 0$ is

$$\rho_s|_{z=0} \quad = \quad \hat{z} \cdot (\mathbf{D}\,|_{z=0^+} - \mathbf{D}\,|_{z=0^-}) = 4\epsilon_o \frac{\mathrm{C}}{\mathrm{m}^2}.$$

c) (3 pts)If the gap is filled with a dielectric of permittivity $\epsilon = 81\epsilon_o$ without changing the surface charge density then the displacement field will remain the same, i.e.,

$$\mathbf{D} = 4\epsilon_o\, \hat{z}\, \frac{\mathrm{C}}{\mathrm{m}^2}.$$

But, the electric field is now

$$\mathbf{E} = \frac{1}{\epsilon} \mathbf{D} = \frac{4}{81}\, \hat{z}\, \frac{\mathrm{V}}{\mathrm{m}}.$$

Consequently, the polarization field becomes

$$\mathbf{P} = \mathbf{D} - \epsilon_o \mathbf{E} = \frac{320}{81} \epsilon_o\, \hat{z}\, \frac{\mathrm{C}}{\mathrm{m}^2}.$$

d) (3 pts)If the medium in the gap has a finite conductivity, then it will also have $\mathbf{E} = 0$ in "steady-state".Thus, $\mathbf{D} \to 0$ and $\mathbf{P} \to 0$. Because, the mobile free charges within the medium in the gap will be pushed and pulled to pile up at the surfaces until the surface charge density generates a secondary field that cancels out the fields wtihin the medium. In this particular case, the salt water shorts out the original field between the plates.

4. The solution of the problem will be given region by region.

- (2 pts)The region defined by $r \leq a$ is occupied by a conductor with $\sigma = 10^6$ S/m, therefore,we can directlywrite
$$\mathbf{D} = \mathbf{0}\,\frac{\mathrm{C}}{\mathrm{m}^2}, \qquad \mathbf{E} = \mathbf{0}\,\frac{\mathrm{V}}{\mathrm{m}}, \qquad \mathbf{P} = \mathbf{0}\,\frac{\mathrm{C}}{\mathrm{m}^2}.$$
for this particular region. In steady-state, charges can accumulate only on the surface of conducting materials. Since this material holds a net charge Q = 4 C, the surface charge density on the sphere of radius r = a is
$$\rho_s|_{r=a} = \frac{Q}{\text{Area}} = \frac{4}{4\pi a^2} = \frac{1}{\pi a^2}\,\frac{\mathrm{C}}{\mathrm{m}^2}.$$

- (3 pts)The region defined by $a < r < b$ is occupied by a dielectric with $\epsilon = 4\epsilon_o$. Applying Gauss's law $\oint \mathbf{D} \cdot d\mathbf{S} = \int_V \rho dV = Q$ and considering $\mathbf{D} = D_r \hat{r}$, we find
$$\mathbf{D} = \frac{4}{4\pi r^2}\hat{r} = \frac{1}{\pi r^2}\hat{r}\,\frac{\mathrm{C}}{\mathrm{m}^2}$$
for $a < r < b$. Hence, the electric field $\mathbf{E}$ is
$$\mathbf{E} = \frac{\mathbf{D}}{\epsilon} = \frac{1}{4\epsilon_0 \pi r^2}\hat{r}\,\frac{V}{\mathrm{m}},$$
from which we can obtain the polarization $\mathbf{P}$ as
$$\mathbf{P} = \mathbf{D} - \epsilon_o \mathbf{E} = (\frac{1}{\pi r^2} - \epsilon_o \frac{1}{4\epsilon_o \pi r^2})\hat{r} = \frac{3}{4\pi r^2}\hat{r}\,\frac{\mathrm{C}}{\mathrm{m}^2}.$$

- (2 pts)As the region defined by $b \leq r \leq c$ has the same material properties as region $r \leq a$, we can write
$$\mathbf{D} = \mathbf{0}\,\frac{\mathrm{C}}{\mathrm{m}^2}, \qquad \mathbf{E} = \mathbf{0}\,\frac{\mathrm{V}}{\mathrm{m}}, \qquad \mathbf{P} = \mathbf{0}\,\frac{\mathrm{C}}{\mathrm{m}^2}.$$
Then, the surface charge density at $r = b$ is given by
$$\begin{aligned} \rho_s|_{r=b} &= \hat{r}.(\mathbf{D}\,|_{r=b^+} - \mathbf{D}\,|_{r=b^-})\,|_{r=b} \\ &= \hat{r} \cdot \left(-\frac{1}{\pi r^2}\hat{r}\right)\Big|_{r=b} = -\frac{1}{\pi b^2}\,\frac{\mathrm{C}}{\mathrm{m}^2}. \end{aligned}$$

- (3 pts)Region defined by $r > c$ is free space. Applying Gauss' law and noting that the total charge enclosed is $4 - 1 = 3\,\mathrm{C}$, we get $\oint \mathbf{D} \cdot d\mathbf{S} = Q = 3\,\mathrm{C}$ where $d\mathbf{S} = \hat{r} 4\pi r^2$. Therefore, we can write
$$\mathbf{D} = \frac{3}{4\pi r^2}\hat{r}\,\frac{\mathrm{C}}{\mathrm{m}^2}, \qquad \mathbf{E} = \frac{3}{4\pi\epsilon_0 r^2}\hat{r}\,\frac{\mathrm{V}}{\mathrm{m}}, \qquad \mathbf{P} = \mathbf{0}\,\frac{\mathrm{C}}{\mathrm{m}^2}.$$
Consequently, the surface charge density at $r = c$ is given by
$$\begin{aligned} \rho_s|_{r=c} &= \hat{r}.(\mathbf{D}\,|_{r=c^+} - \mathbf{D}\,|_{r=c^-})\,|_{r=c} \\ &= \hat{r} \cdot \left(\frac{3}{4\pi r^2}\hat{r}\right)\Big|_{r=c} = \frac{3}{4\pi c^2}\,\frac{\mathrm{C}}{\mathrm{m}^2}. \end{aligned}$$

5. a)

i.

V due to Q, we can get: $$V_+(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{|\mathbf{r}^+|}$$

V due to -Q, we can get: $$V_-(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_0}\frac{1}{|\mathbf{r}^-|}$$

Then, according to the superposition principle, we obtain:

$$V(\mathbf{r}) = V_+(\mathbf{r}) + V_-(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_0}(\frac{1}{|\mathbf{r}^+|} - \frac{1}{|\mathbf{r}^-|}) = \frac{Q}{4\pi\varepsilon_0}(\frac{|\mathbf{r}^-| - |\mathbf{r}^+|}{|\mathbf{r}^+||\mathbf{r}^-|})$$

ii.

We bring the $r^+ \to r - \frac{d}{2}\cos\theta, r^- \to r + \frac{d}{2}\cos\theta$

$$V(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_0}(\frac{d\cos\theta}{r^2 - (\frac{d}{2}\cos\theta)^2})$$

Due to $r \ggg d$, we obtain:

$$V(\mathbf{r}) = \frac{Q}{4\pi\varepsilon_0}(\frac{d\cos\theta}{r^2})$$

b)

i.

Due to $V_{in}(\mathbf{r}) = A\hat{z}$, we can obtain:

$$\mathbf{E_{in}} = -\nabla V_{in}(\mathbf{r}) = -A\hat{z}$$

From the definition of $\mathbf{D}$,we can get:

$$\mathbf{D}_{in} = \varepsilon_0\mathbf{E}_{in} + \mathbf{P} = -A\varepsilon_0\hat{z} + P\hat{z} = (P - A\varepsilon_0)\hat{z}$$

ii.

Due to $V_{out}(\mathbf{r}) = B\frac{\cos\theta}{r^2}$ and $\nabla V = \hat{r}\frac{\partial V}{\partial t} + \hat{\theta}\frac{1}{r}\frac{\partial V}{\partial \theta} + \hat{\phi}\frac{1}{r\sin\theta}\frac{\partial V}{\partial \phi}$, we can obtain:

$$\mathbf{E_{out}} = -\nabla V_{out} = \hat{r}\frac{\partial V_{out}}{\partial t} + \hat{\theta}\frac{1}{r}\frac{\partial V_{out}}{\partial \theta} = \hat{r}\frac{2B\cos\theta}{r^3} + \hat{\theta}\frac{B\sin\theta}{r^3}$$

$$\mathbf{D_{out}} = \varepsilon_0\mathbf{E}_{out} = \hat{r}\frac{2B\varepsilon_0\cos\theta}{r^3} + \hat{\theta}\frac{B\varepsilon_0\sin\theta}{r^3}$$

iii.

①

Due to $V_{in}(r = a) = V_{out}(r = a)$, $V_{in}(r = a) = A\hat{z}|_{r=a} = Aa\cos\theta$ and $V_{out}(r = a) = B\frac{\cos\theta}{r^2}|_{r=a} = B\frac{\cos\theta}{a^2}$,we can obtain:

$$V(a) = B\frac{cos\theta}{a^2} = Aacos\theta$$

$$B = Aa^3$$

②

According to the boundary conditions, we obtain:

$$\hat{r}\cdot(\mathbf{D}_{out} - \mathbf{D}_{in}) = 0$$

From the i. and ii. , we can get:

$$\mathbf{D}_{in} = (P - A\varepsilon_0)\hat{z} = (P - A\varepsilon_0)$$

$$\mathbf{D_{out}} = \hat{r}\frac{2B\varepsilon_0 \cos\theta}{r^3} + \hat{\theta}\frac{B\varepsilon_0 \sin\theta}{r^3}$$

Then, we can get:

$$\hat{r} \cdot (\mathbf{D}_{out} - \mathbf{D}_{in}) = \frac{2\varepsilon_0 B cos\theta}{a^3} - (P - \varepsilon_0 A)cos\theta = 0$$

So,

$$A = \frac{P}{3\varepsilon_0}$$
$$B = \frac{a^3 P}{3\varepsilon_0}$$

$$\mathbf{E}_{in} = -A\hat{z} = -\frac{P}{3\varepsilon_0}\hat{z} = -\frac{\mathbf{P}}{3\varepsilon_0}$$

iv.
According to the boundary conditions, we obtain:

$$\mathbf{n} \times (\mathbf{E}_{out} - \mathbf{E}_{in}) = 0$$

Among,

$$\mathbf{E_{in}} = -A\hat{z}$$

$$\mathbf{E_{out}} = \hat{r}\frac{2B\cos\theta}{r^3} + \hat{\theta}\frac{B\sin\theta}{r^3}$$

$$\hat{z} = \hat{r}\cos\theta - \hat{\theta}\sin\theta$$

We can get:

$$\mathbf{E}_{in,t} = \mathbf{E}_{out,t}$$

$$\mathbf{E}_{in,t} = Asin\theta$$

$$\mathbf{E}_{out,t} = \frac{Bsin\theta}{a^3}$$

Therefore,

$$Asin\theta = \frac{Bsin\theta}{a^3}$$

e.g. $B = a^3 A$ equals to **iii** ①, so they are equiralent.

6 **Continuity equation and transient response in conductors**

a) (2 pts.) Using Gauss's law to replace the second term in the continuity equation as:

$$\nabla \cdot \mathbf{J} = \nabla \cdot (\sigma \mathbf{E}) = \sigma \nabla \cdot \mathbf{E} = \frac{\sigma}{\epsilon_0}\rho,$$

and we have the differential equation for the charge density ρ:

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon_0}\rho = 0. \tag{1}$$

b) (2 pts.) We know the solution to (1) take the form

$$\rho(t) = \rho_0 e^{-\frac{\sigma}{\epsilon_0}t},$$

where ρ_0 is some constant to be determined. Using the given condition at $t = 0$, we get

$$\rho(x, y, z, t) = \cos(5x)e^{-\frac{\sigma}{\epsilon_0}t}\,\mathrm{C/m^3}. \tag{2}$$

c) (2 pts.) Using (2), we can find the time t for the charge density to reduce to $0.01\cos(5x)\,\mathrm{C/m^3}$ from

$$\rho(x, y, z, t) = \cos(5x)e^{-\frac{\sigma}{\epsilon_0}t} = 0.01\cos(5x),$$

which gives $t = \frac{\epsilon_0}{\sigma}\ln(100) = 7.03 \times 10^{-19}\,\mathrm{s}$.

d) (2 pts.) Since the electric field only has an x-component with x-dependency, we have

$$\rho(x, t) = \cos(5x)e^{-\frac{\sigma}{\epsilon_0}t} = \nabla \cdot \mathbf{D},$$

then using the constitution relation $\mathbf{D} = \epsilon_0 \mathbf{E}$, we have

$$\nabla \cdot \mathbf{D} = \nabla \cdot (\epsilon_0 \mathbf{E}) = \epsilon_0 \nabla \cdot \mathbf{E} = \epsilon_0 \frac{\partial E_x}{\partial x} = \cos(5x)e^{-\frac{\sigma}{\epsilon_0}t}. \tag{3}$$

The anti-derivative of (3) gives the solution to the electric field as

$$\mathbf{E}(x, y, z, t) = \hat{x}\frac{1}{5\epsilon_0}\sin(5x)e^{-\frac{\sigma}{\epsilon_0}t}\,\mathrm{V/m}. \tag{4}$$

e) (2 pts.) By inspecting (4), we know the electric field goes to 0 when $t \to \infty$. Since $\mathbf{J} = \sigma \mathbf{E}$, we conclude that $\mathbf{J} \to 0$ as $t \to \infty$.

7. **Capacitance, conductance - practical case**

a) (2.5 pts.) The capacitance of the capacitor will be

$$C = \frac{\epsilon A}{d} = \frac{\epsilon_o \pi (2 \times 10^{-2})^2}{5 \times 10^{-3}} \approx 2.23\,\mathrm{pF}.$$

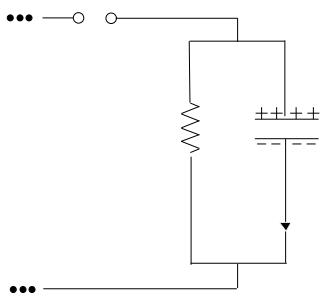
b) (2.5 pts.) The conductance of the same capacitor will be

$$G = \frac{\sigma A}{d} = \frac{10^{-3}(2 \times 10^{-2})^2}{5 \times 10^{-3}} = 8 \times 10^{-5}\,\mathrm{S},$$

whereas corresponding resistance is

$$R = \frac{1}{G} = 12.5\,\mathrm{k\Omega}.$$

c) (2.5 pts.) You can see the equivalent circuit of the capacitor in the following figure when the switch is open. As you can see, the current direction is reversed.



Using Kirchoff's voltage rule, we can write the voltages in the loop as

$$IR + \frac{Q}{C} = 0,$$

which yields

$$R\frac{dQ}{dt} + \frac{Q}{C} = 0$$

after replacing I with $\frac{dQ}{dt}$. Solution to the above first-order differential equation is given by

$$Q(t) = Q_o e^{-t/RC},$$

where RC in the exponential terms denotes the time constant. Time constant τ of the circuit is $\tau = RC = 12.5 \times 10^3 \times 2.23 \times 10^{-12} = 27.875\,\text{ns}$. Finally, we find the charge on the capacitor at $t = 30\,\text{ns}$

$$Q(t = 30\,ns) = 1e^{-30/27.875} \approx 0.34\,\text{C}.$$

d) (2.5 pts.) Since we have $t \gg R_{th}C$, using the voltage division rule, we can write

$$V(t = 20\,s) = 5\frac{12.5}{(12.5 + 1)} = 4.63\,\text{V}.$$

8. **Spherical capacitor:**

a) (2 pts.) Applying the integral form of Gauss's law, i.e. $\epsilon \oint \mathbf{D} \cdot d\mathbf{S} = Q_\text{V}$ to a sphere of radius r (where $a < r < b$), we get

$$4\pi r^2 E_r = \frac{Q}{\epsilon},$$

from which we find

$$\mathbf{E} = E_r\hat{r} = \frac{Q}{4\pi\epsilon r^2}\hat{r}.$$

Taking the path integral of the electric field $\mathbf{E}$ from a to b, i.e. $V = -\int_a^b \mathbf{E} \cdot d\mathbf{l}$,

$$\begin{aligned} V_{ab} &= V_a - V_b = -\int_b^a \mathbf{E} \cdot \hat{r}dr \\ &= \int_a^b \mathbf{E} \cdot \hat{r}dr = \int_a^b \frac{Q}{4\pi\epsilon r^2}dr, \end{aligned}$$

we find the potential drop V from the inner to outer shell as

$$V = \frac{Q}{4\pi\epsilon}(\frac{b-a}{ab}).$$

Since $C = \frac{Q}{V}$, we obtain

$$C = 4\pi\epsilon(\frac{ab}{b-a}).$$

b) (2 pts.) Referring to the result we have found in part (a), the capacitance C is given by

$$C = 8\pi\epsilon_o \frac{a}{(1-\frac{a}{b})}.$$

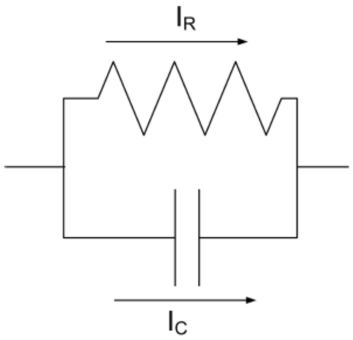
Given that $b \to \infty$ and $a = 1\,\text{m}$, we find

$$C = 8\pi\epsilon_o \approx 222.53\ \text{pF}.$$

c) (2 pts.) Using the formula given in the problem, the conductance G is given

$$\begin{aligned} G &= \frac{\sigma}{\epsilon} C \text{by} \\ &= \frac{2\times 10^{-6}}{2\times 8.85\times 10^{-12}} \times 222.53\times 10^{-12} \\ &\approx 27.81\,\mu\text{S}. \end{aligned}$$

d) (2 pts.) The RC equivalent circuit that represent the physical system described above is a parallel R and C circuit as shown in the below figure.



Here, resistance $R = \frac{1}{G}$ is in parallel with capacitance C. The two terminals represent the surface of the spherical conducting shell and a point at infinity. Because the voltage across both elements is the same (equal to the voltage between the shell and infinity), they must be in parallel. In the given situation, no source is connected across the terminals, only an initial voltage is present, caused by the initial charge on the shell. The charge on the capacitor is $Q = CV$ and the equation describing the current flow can be rewritten in terms of the charge on the shell as follows. We start applying KCL on the node on the left,

$$I_R + I_C = 0,$$

$$GV + C\frac{\partial V}{\partial t} = 0,$$

$$\frac{\partial Q}{\partial t} + \frac{GQ}{C} = 0.$$

e) (2 pts.) Initial condition for the differential equation in part (d) is that $Q(t) = 1\,\mathrm{C}$ on the inner shell at $t = 0$. To obtain an expression for $Q(t)$ in $t > 0$, we will refer to the equation we have found in part (d),

$$\frac{\partial Q}{\partial t} + \frac{GQ}{C} = 0,$$

which yields a general solution

$$Q(t) = a_1 e^{-\frac{G}{C}t} + a_2.$$

The initial condition can be written as $Q(0) = 1\,\mathrm{C}$ and at $t \to \infty$ we expect no charge $Q(\infty) = 0$. These conditions lead to $a_2 = 0$ and $a_1 = 1\,\mathrm{C}$. Therefore, the final expression will be

$$Q(t) = e^{-(\frac{G}{C}t)}\,\mathrm{C}.$$

9. **Copper wire**

a) (2.5 pts.) Using the formula given in the problem and Lecture 10, the resistance R of this copper wire is calculated as

$$R = \frac{d}{A\sigma} \approx 1.098\,\Omega.$$

b) (2.5 pts.) The voltage drop across the wire is given by $V = E \cdot d = R \cdot I$, where I is the current flowing along the wire. Assuming $I = 1\,\mathrm{A}$, the electric field within the wire is

$$E = \frac{R \cdot I}{d} \approx 5.488 \times 10^{-3}\,\frac{\mathrm{V}}{\mathrm{m}}.$$

c) (2.5 pts.) Referring to the hint given in the problem, the current density may be expressed as $\mathbf{J} = \sigma\mathbf{E} = N_e q_e \mathbf{v}_e$ where $q_e = -1.6 \times 10^{-19}\,\mathrm{C}$. Then, the mean speed of an electron is given by

$$|\mathbf{v}_e| = \frac{\sigma\,|\mathbf{E}|}{N_e\,|q_e|} \approx 2.354 \times 10^{-5}\,\frac{\mathrm{m}}{\mathrm{s}}.$$

d) (2.5 pts.) The time it would take an electron to travel from one end of the wire to the other is simply

$$t = \frac{d}{|\mathbf{v}_e|} \approx 8.495 \times 10^{6}\,\mathrm{s}.$$